

Catalytic Impact of Early Investments in Advanced Clean Energy Technologies

Kick-off

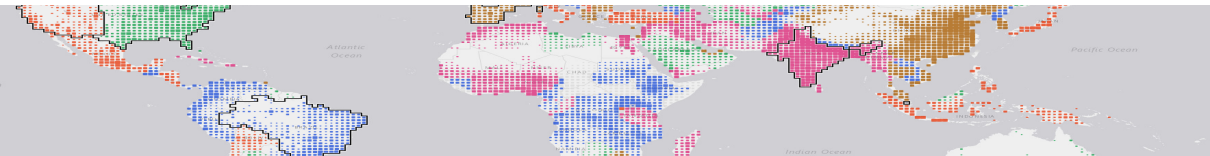
PyPSA Labs, Google

8 September 2026



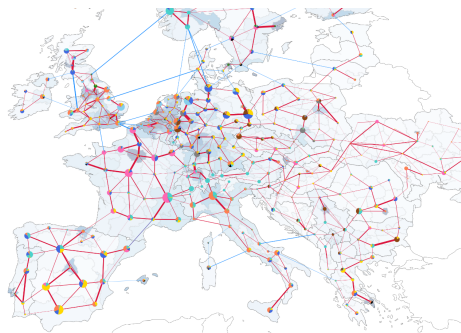
Project overview

- Lay the foundation for understanding the **catalytic impact of early investments** in advanced clean electricity technologies
- Quantitative illustration of the **direct and indirect** system-wide emissions reductions and cost-decline trajectories
- Inform **consequential and out-of-scope emissions accounting methodologies** and the broader conversation on how early-mover investments should be valued
- Deliver an **open-source, reproducible scientific workflow**



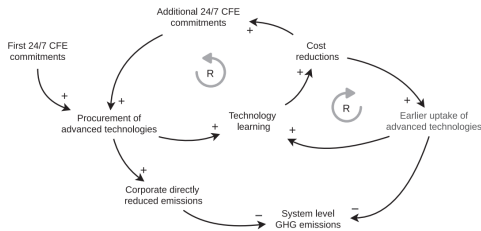
Modelling setup 1/2

- **PyPSA** for capacity expansion and dispatch optimisation
- **PyPSA-Earth** as the global data pipeline: brownfield fleet, demand, grid topology
- **atlite** for renewable resource potentials and capacity factors
- Representative regions *aka grid archetypes* featuring jurisdictions with varying grid topologies, renewable resource potentials, technology portfolios, etc.



Modelling setup 2/2

- Use 2025 as a **validation year**
- Run a **capacity-expansion model** for each grid archetype across investment years from 2025 to 2050
- Clear markets for advanced technologies between investment years; learning curves pass updated costs forward **myopically**
- Compare a **benchmark scenario** without catalytic action with a scenario including action by early investors
- Like-for-like comparison in system metrics (e.g. avoided emissions) to quantify impact of catalytic action



Scenarios

S1 — Reference pathway

Each archetype follows current cost trajectories and announced retirements, without catalytic investment. **Establishes the counterfactual baseline** for all impact estimates.

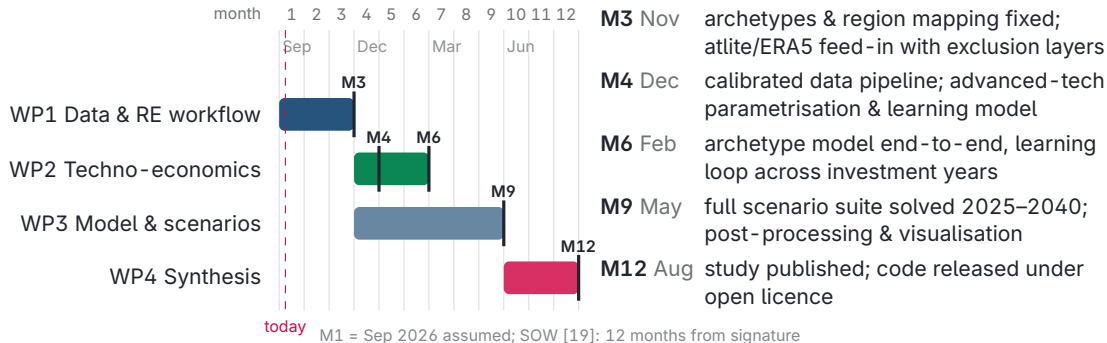
S2 — Catalytic early investment

Corporate and institutional buyers directly procure advanced clean technologies, with investment scale anchored to a share of regional demand and the portfolio optimised for local system needs. **Early deployment accelerates learning and cost competitiveness** beyond buyers' regions.

Additional scenarios represent alternative futures to baseline.

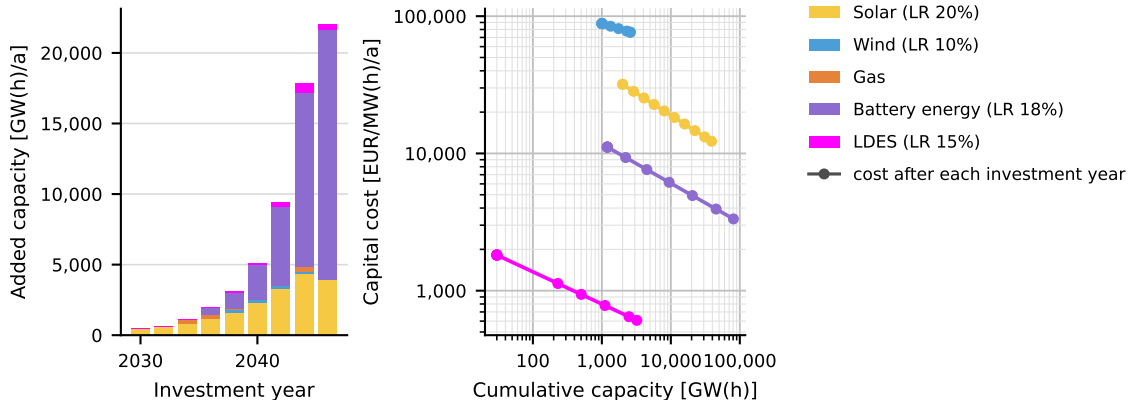
- **Wind/solar acceptance limits:** tighter land-use, protected-area and social-acceptance constraints increase the value of clean firm power.
- **CDR breakthrough:** engineered removal reaches about 150 EUR/tCO₂ by 2040, making gas plus CDR a competitive backup option.
- **Advanced baseload breakthrough:** rapid FOAK-to-commercial cost decline for SMRs or another firm clean technology.
- **Advanced geothermal breakthrough:** commercial success in geologically suitable regions creates locally significant catalytic impact.
- **SOFC breakthrough:** enable cheap, efficient backup running on hydrogen or natural gas with high CO₂ capture; reduces the premium for LDES and flexible clean generation.

SOW work packages and milestones



Prototype: Endogenous learning in the world model

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Open questions

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- Technology palette
- Initial investment mechanism
- Modelled regions

Technology palette

Mature: exogenous costs

- Solar PV, onshore/offshore wind
- Hydro: reservoir, run-of-river, pumped hydro
- Biomass
- Nuclear (large LWR)
- Gas CCGT and OCGT
- Li-ion battery

Advanced: endogenous learning

- LDES: CO₂ battery, iron-air battery, H₂ + SOFC, vanadium flow
- Nuclear SMR
- Next-gen geothermal: EGS, closed-loop
- Allam-cycle gas + CCS
- CDR: direct air capture, cost cut-off

The Initial Investment Mechanism



- Generalise the initial action beyond 24/7 CFE to represent **multiple forms of catalytic investment**
- Let the model select the **locally appropriate technology portfolio**, rather than prescribing which technologies are procured
- Test two mechanisms with endogenous portfolio selection: a **demand-linked procurement target** and a **green-premium budget**

Modelled regions and grid archetypes

Archetypes proposed in the SOW

- **Renewables - constrained, high - density:** Western Europe, South Korea, Japan
- **Geothermal - favourable:** US West, South - East Asia, Turkey, Italy
- **Islanded / weakly interconnected:** Western Australia, Ireland
- **Hydro - rich:** Norway, Brazil, parts of Canada
- **Fossil - heavy brownfield with large economies:** parts of the US and Europe, India

Archetypes and regions selected



Renewables - constrained, high - density

☐ modelled: North-West Europe



Geothermal - favourable

☐ modelled: United States (West)



Islanded / weakly interconnected

☐ modelled: Singapore



High renewables potential

☐ modelled: India



Hydro-rich

☐ modelled: Brazil

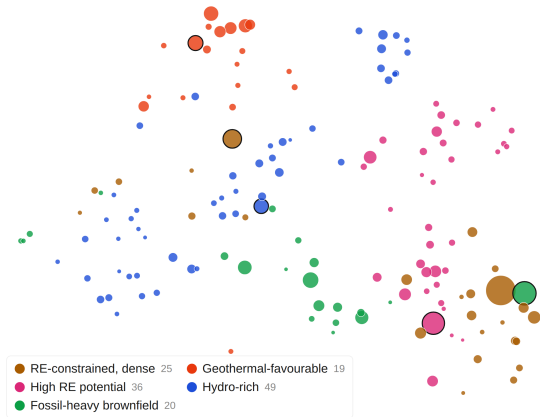


Fossil-heavy brownfield with large economies

☐ modelled: United States (East)

k -means for archetype classification

- **Autarky:** 1 at zero trade, 0 at $|\text{net imports}| \geq 20\%$ of demand
- **Population density:** $\log_{10}$ people per km^2
- **2050 demand per capita:** $\log_{10}$ kWh per person
- **Demand growth to 2050:** $\log_2$ of 2050 over today's demand
- **Clean incumbent share:** hydro + nuclear, % of generation
- **Fossil share:** coal, gas and oil, % of generation
- **Geothermal potential:** installed capacity, EGS-suitable land, temperature at 5 km



Modelled regions and grid archetypes



Renewables-constrained, high-density

□ modelled: North-West Europe



Geothermal-favourable

□ modelled: United States (West)



Islanded / weakly interconnected

□ modelled: Singapore



High renewables potential

□ modelled: India



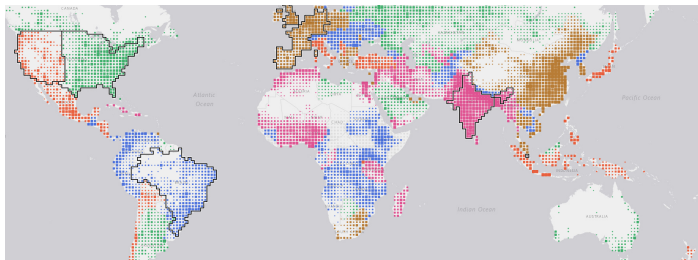
Hydro-rich

□ modelled: Brazil



Fossil-heavy brownfield with large economies

□ modelled: United States (East)



Modelled regions and grid archetypes



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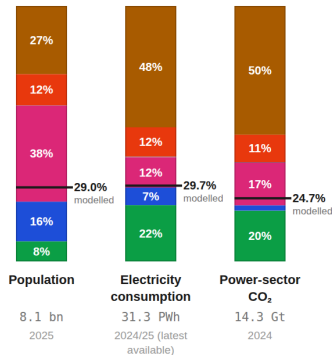
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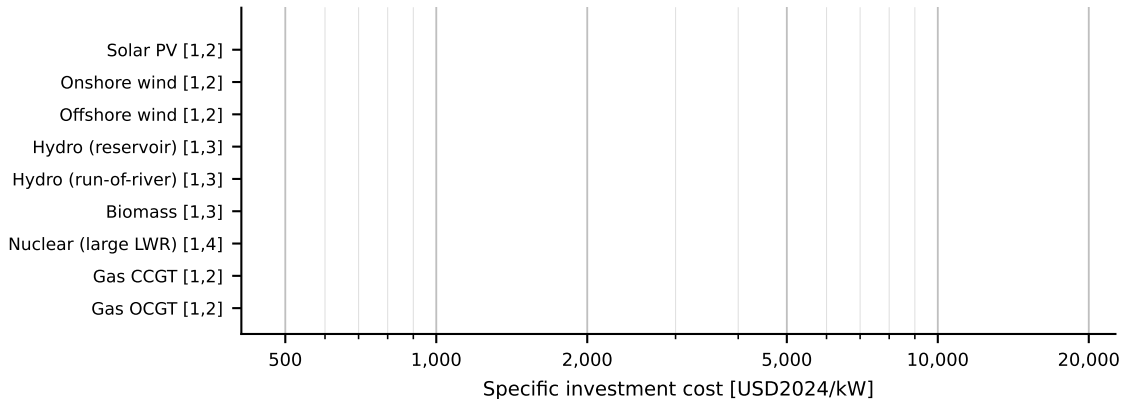
Fossil-heavy brownfield with large economies

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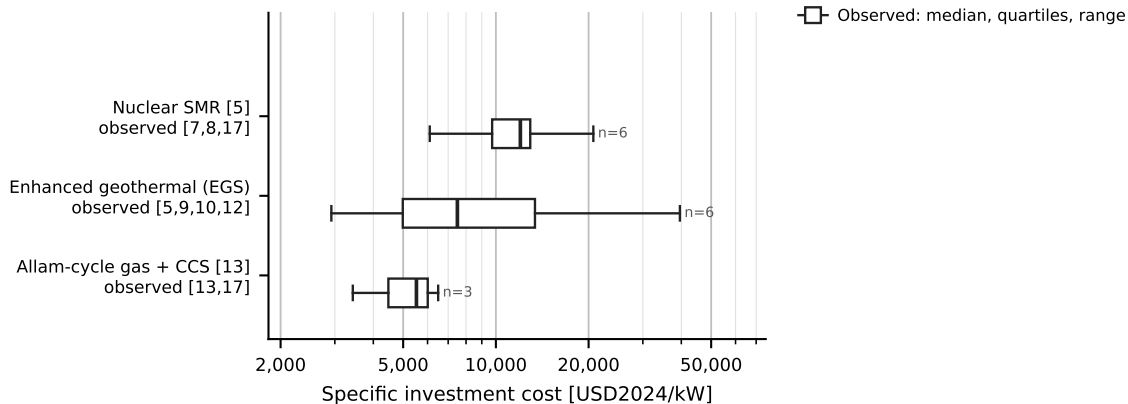


Annex

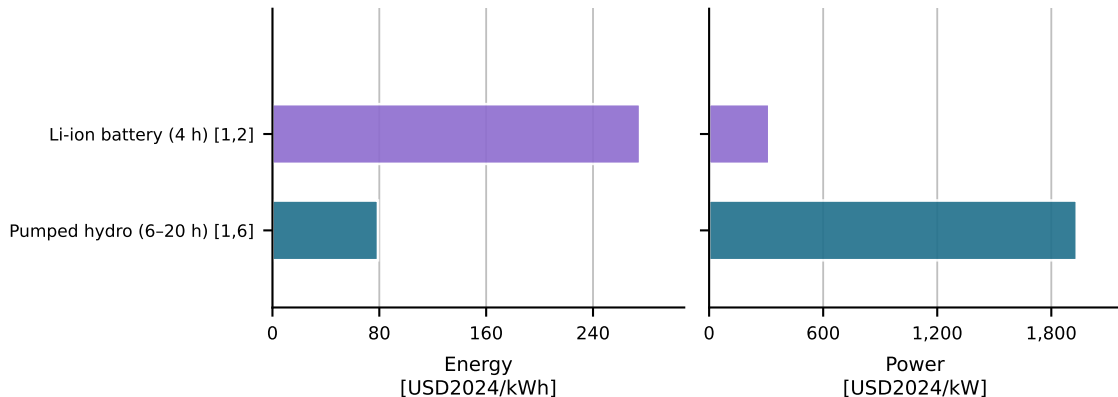
2025 capex baseline: mature generation



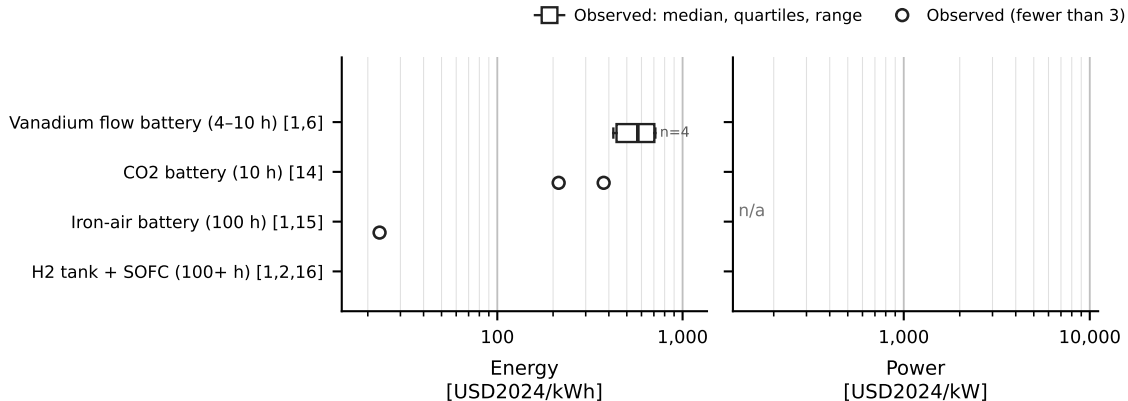
2025 capex baseline: advanced generation



2025 capex baseline: mature storage



2025 capex baseline: advanced storage



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Get in touch

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